

Sleep Health and Lifestyle Analysis

A Data-Driven Project Report

Project Duration: June 2026

Dataset: Sleep Health and Lifestyle (374 records)

Tools: Python (pandas, matplotlib, seaborn), Microsoft PowerPoint

Executive Summary

This project analyzes the relationship between sleep quality and lifestyle factors using a dataset of 374 individuals. The key finding is that sleep quality is NOT uniformly proportional to all lifestyle indicators—it is strongly tied to psychological and physiological factors (stress, heart rate, sleep duration), but only weakly tied to physical activity and step count.

- **Sample size:** 374 individuals with complete lifestyle and sleep health data
- **Primary finding:** Stress shows the strongest inverse correlation ($r = -0.90$) with sleep quality
- **Secondary finding:** 41.4% of individuals report a sleep disorder (Insomnia or Sleep Apnea)
- **Data quality:** Zero missing values after cleaning; all 374 records complete
- **Methodology:** Exploratory data analysis, Pearson correlation, statistical visualization

1. Introduction and Objectives

Sleep quality is fundamental to overall health, but its relationship to lifestyle factors is complex and often misunderstood. Many people assume that simply increasing physical activity or reducing stress will automatically improve sleep, but this project tests whether that assumption holds true in the data.

1.1 Project Objectives

1. Understand the dataset structure and explore the distribution of sleep quality across 374 individuals.
2. Clean and preprocess the data to ensure data integrity and consistency.
3. Analyze lifestyle factors (stress, physical activity, BMI, occupation) and their impact on sleep quality.
4. Quantify the strength of relationships using Pearson correlation analysis.
5. Test the hypothesis: Is sleep quality directly proportional to lifestyle factors?

2. Dataset Description

2.1 Source and Structure

The dataset comprises 374 records with 13 original attributes covering sleep metrics, lifestyle factors, demographics, and health indicators. Each record represents a distinct individual with complete data across all variables.

Attribute	Description
Total Records	374 individuals
Original Attributes	13 columns (expanded to 15 after cleaning)
Missing Values	0 (fully complete)

2.2 Key Variables

- **Sleep Metrics:** Sleep Duration (hours), Quality of Sleep (1–10 scale)
- **Lifestyle:** Physical Activity Level (scale), Stress Level (1–10), Daily Steps, Occupation
- **Health:** BMI Category (Normal, Overweight, Obese), Blood Pressure, Heart Rate
- **Outcome:** Sleep Disorder (No Disorder, Insomnia, Sleep Apnea)
- **Demographics:** Age, Gender

3. Data Cleaning and Preprocessing

Although the dataset was relatively clean, several preprocessing steps were applied to ensure data consistency and prepare it for analysis.

3.1 Cleaning Steps Applied

6. **Sleep Disorder missing values:** 219 blank entries were explicitly filled as 'No Disorder' (not dropped), representing individuals without a diagnosed sleep disorder.
7. **BMI Category inconsistency:** 'Normal' and 'Normal Weight' were identified as the same category and merged into a single 'Normal' label.
8. **Blood Pressure splitting:** The combined text field 'SystolicBP/DiastolicBP' (e.g., '120/80') was split into two numeric columns for statistical analysis.
9. **Duplicate check:** 242 records share identical lifestyle/health profiles (excluding Person ID); these were retained as they represent distinct individuals.
10. **Data type validation:** All numeric columns confirmed as numeric; date fields confirmed as text.

3.2 Outcome

Final cleaned dataset: 374 rows × 15 columns (2 new numeric columns created from Blood Pressure split). Zero missing values. All records retained.

4. Exploratory Data Analysis

4.1 Descriptive Statistics

Key summary statistics for numeric variables:

Variable	Mean	Std Dev	Min	Max
Sleep Duration	7.13 hrs	0.80	5.8 hrs	8.5 hrs
Quality of Sleep	7.31	1.20	4.0	9.0
Stress Level	5.39	1.77	3.0	8.0

4.2 Demographic Distribution

- **Gender:** Male 50.5% (189), Female 49.5% (185)—well-balanced sample
- **BMI Distribution:** Normal 57.8% (216), Overweight 39.6% (148), Obese 2.7% (10)
- **Sleep Disorder Prevalence:** No Disorder 58.6% (219), Sleep Apnea 20.9% (78), Insomnia 20.6% (77)
- **Age Range:** 27–59 years (Mean: 42.18, SD: 8.67)

5. Correlation Analysis: Testing Proportionality

The central research question is whether sleep quality is directly proportional to lifestyle factors. To answer this, Pearson correlation coefficients were calculated between Quality of Sleep and nine lifestyle/health variables. The results reveal nuanced relationships:

Lifestyle Factor	r-value	Strength
Stress Level		Very Strong
Sleep Duration	+0.88	Very Strong
Heart Rate	-0.66	Moderate
Physical Activity Level	+0.19	Weak
Daily Steps	+0.02	None

5.1 Interpretation

The data reveals a dichotomy: Sleep quality is **STRONGLY** proportional to stress and sleep duration ($r > |0.88|$), but **NOT** proportional to activity-based metrics ($r \approx 0.19$ and 0.02). This suggests that psychological stress and physiological sleep metrics are far more predictive of sleep quality than the amount of physical activity or daily steps.

Notably, the near-zero correlation with Daily Steps contradicts a common assumption that 'more steps = better sleep.' Sleep quality is context-dependent and driven more by stress management and adequate sleep duration than by exercise volume alone.

6. Key Findings and Insights

6.1 Stress Is the Strongest Sleep Predictor

With a correlation of $r = -0.90$, stress level is the single strongest predictor of sleep quality in this dataset. As stress increases by one unit (on a 1–10 scale), sleep quality tends to decrease sharply and predictably. This finding aligns with well-established research on the sleep–stress axis: elevated cortisol and mental tension directly suppress sleep onset and degrade sleep architecture.

6.2 Sleep Duration and Quality Move in Lockstep

Sleep Duration and Quality of Sleep are highly correlated ($r = 0.88$), suggesting both metrics capture overlapping aspects of sleep health. Individuals who sleep longer tend to self-report higher quality, and vice versa. However, the correlation is not perfect ($r < 1$), indicating that duration and quality are distinct dimensions—one can have long sleep of poor quality, and short sleep of high quality.

6.3 Occupation and Sleep Quality

Sleep quality varies noticeably by occupation (range: 4.0–8.41):

- **Best:** Engineers (8.41), Lawyers (7.89), Accountants (7.89)
- **Worst:** Sales Representatives (4.0), Scientists (5.0)
- Note: Sales and Scientist groups have low sample sizes (2 and 4 records), so treat as indicative rather than conclusive. This pattern suggests that job-related stress and schedule flexibility are indirect drivers of sleep quality.

6.4 BMI Category and Sleep Quality

Sleep quality deteriorates with increasing BMI category: Normal (7.64) > Overweight (6.90) > Obese (6.40). This 1.24-point decline across BMI categories reinforces that overall physical health (proxied by BMI) is linked to sleep outcomes. The pattern may reflect that excess weight contributes to sleep apnea, discomfort, and compromised sleep architecture.

6.5 Sleep Disorder Distribution

41.4% of the sample reports a diagnosed sleep disorder. Insomnia and Sleep Apnea are nearly equally prevalent (20.6% and 20.9%, respectively). Individuals with Insomnia show the lowest sleep quality (6.53) and highest stress (5.87), suggesting that stress-induced insomnia is a

dominant disorder type. Sleep Apnea individuals show slightly higher quality (7.21) despite the diagnosis, possibly due to treatment or that their apnea is detected but not yet severely impacting perceived quality.

6.6 Additional Insights from Extended Analysis

- **Heart Rate & Stress:** Strong positive correlation ($r = 0.67$), indicating that stress elevates resting heart rate, which in turn is associated with poorer sleep.
- **Age & Sleep Quality:** Weak positive correlation ($r = 0.47$); older individuals in this sample tend to report slightly higher sleep quality, contrary to some literature suggesting age-related sleep decline.
- **Gender Differences:** Nearly equal representation (50.5% male, 49.5% female). Sleep quality and disorder prevalence show minimal gender difference in univariate analysis.

7. Conclusion

This analysis examined whether sleep quality is directly proportional to lifestyle factors. The honest answer is: PARTIALLY.

Sleep quality is STRONGLY proportional to: Stress level ($r = -0.90$), Sleep Duration ($r = 0.88$), and Heart Rate ($r = -0.66$). These psychological and physiological factors are powerful predictors.

Sleep quality is WEAKLY proportional to: Physical Activity Level ($r = 0.19$) and Daily Steps ($r = 0.02$). Exercise volume alone is not a strong determinant of sleep quality.

This finding has practical implications: improving sleep quality requires a holistic approach—managing stress through relaxation techniques, maintaining regular sleep schedules, and achieving overall physical health are more effective than simply increasing exercise. The data suggests that exercise is beneficial for sleep, but not as a standalone intervention.

8. Future Scope

This project opens several avenues for future investigation:

11. Build predictive models (regression, random forest) to forecast sleep quality from lifestyle inputs and compare model performance.
12. Expand dataset with wearable device data (REM sleep, deep sleep, sleep onset latency) to refine understanding of sleep architecture.
13. Conduct longitudinal analysis tracking individuals over time to test causality, not just correlation.
14. Investigate interaction effects (e.g., does stress management reduce the impact of high BMI on sleep?).
15. Segment the population by disorder type and test whether relationships differ between Insomnia and Sleep Apnea subgroups.
16. Develop a clinical intervention framework based on the correlation hierarchy—prioritizing stress reduction and sleep schedule optimization over activity targets.

9. Appendix: Methodology and Tools

9.1 Analysis Tools

- **Language:** Python 3.10+
- **Data Processing:** Pandas, NumPy
- **Statistical Methods:** Pearson correlation, descriptive statistics
- **Visualization:** Matplotlib, Seaborn (13 publication-quality charts)

9.2 Charts Generated

17. Sleep Duration & Quality of Sleep Distributions
18. Quality of Sleep vs Stress Level (box plot)
19. Quality of Sleep vs Physical Activity Level (scatter)
20. Sleep Duration vs Daily Steps (regression)
21. Average Sleep Quality by Occupation (bar chart)
22. Sleep Quality by BMI Category (box plot)
23. Correlation Heatmap (9 variables)
24. Sleep Disorder Prevalence by BMI (stacked bar)
25. Gender & Sleep Disorder Distribution (pie charts)
26. Average Sleep Duration by Sleep Disorder (bar)
27. Age Distribution by Sleep Disorder (violin plot)
28. Sleep Quality by Gender and Sleep Disorder (grouped bar)
29. Heart Rate vs Stress Level (regression)

9.3 Statistical Notes

- **Correlation Strength:** $r > |0.7|$ = Very Strong; $|0.5|$ to $|0.7|$ = Strong; $|0.3|$ to $|0.5|$ = Moderate; $< |0.3|$ = Weak
- **Causality:** Correlations imply association, not causation. Direction of causality cannot be inferred from this cross-sectional data.
- **Sample Bias:** Sales Representatives ($n=2$) and Scientists ($n=4$) have small sample sizes and should be interpreted cautiously.

END OF REPORT